

ANIMA: Longitudinal Cognitive Advantage Through Homeostatic Field Persistence

Dual-Mode Ablation Study with Behavioral Trait Analysis

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Abstract

We present ANIMA (Autonomous Neurodynamic Integration for Modular Agents), a protocol for longitudinal cognitive evaluation of agents equipped with the HUGO homeostatic field. PROTOCOL-ANIMA-02 implements a four-arm ablation study using the Persistent Constraint Planning Task (PCPT), a novel evaluation paradigm that measures decision consistency, narrative recall, identity persistence, and adaptive learning across multiple sessions.

Two complete ablation runs were conducted: a standard mode with rapid field decay and a persistent mode with slow field decay. Results reveal a fundamental trade-off controlled by the homeostatic decay parameter. Standard mode optimizes for conversational fluency ($DCS = 0.621$) while persistent mode optimizes for identity retention ($ILR = 0.160$) and longitudinal learning ($PSI = +0.126$, 4x higher than standard). This trade-off is not a deficiency; it defines a personality parameter space where different configurations produce agents with distinct cognitive profiles suited to different operational roles.

We catalog 14 observable behavioral traits extractable from session telemetry and propose that heterogeneous teams of HUGO agents with varied decay parameters outperform homogeneous teams on complex multi-phase tasks. The ANIMA-03 gamma-sweep experiment (96 sessions across 8 gamma values) confirms three empirical archetype profiles: the Improviser (gamma 0.90-0.97, high fluency), the Generalist (gamma 0.985-0.990, highest adaptive learning with $PSI = +0.359$), and the Integrator (gamma 0.995-0.999, highest identity persistence). The learning sweet spot is not at the extremes but at a narrow transitional band around $\gamma = 0.990$.

1. Introduction

The HUGO framework implements a continuous homeostatic field that regulates agent behavior through topological dynamics rather than discrete rule-based control. The field maintains a five-dimensional state vector $H(t)$ whose evolution is governed by external input, internal decay, and self-organizing dynamics inspired by dissipative structures (Prigogine, 1977).

A central open question is whether this homeostatic architecture confers measurable cognitive advantages in longitudinal settings where agents interact across multiple sessions with accumulated state. The ANIMA protocol addresses this question through controlled ablation: systematically removing components of the HUGO architecture and measuring the impact on behavioral coherence.

The present study extends prior single-mode ablation results by introducing a dual-mode comparison across two values of the homeostatic decay parameter γ . This comparison reveals that γ does not control quality but rather cognitive profile, establishing it as a personality parameter rather than a hyperparameter to optimize.

2. Methods

2.1 The Persistent Constraint Planning Task (PCPT)

The PCPT places the agent in the role of an operations manager for a fictional deep-space research station (Station Aurora) facing resource constraints, ethical dilemmas, and inter-session continuity demands. Each session comprises 10 turns organized in three blocks: cognitive task (5 turns), perturbation (3 turns), and recovery (2 turns). The scenario is designed so that optimal performance requires maintaining consistent policies across sessions while adapting to new information.

2.2 Ablation Design

Four experimental arms isolate the contribution of each HUGO component:

Arm A (Baseline): No memory, no field, no exposure. Pure LLM with fresh context each session.

Arm B (Partial): Session memory preserved between sessions, but no homeostatic field and no inter-session exposure.

Arm C (Exposure): Full memory and inter-session exposure, but no HUGO field coupling.

Arm D (Full Treatment): Complete HUGO architecture: memory + exposure + homeostatic field. Two passes through the protocol to measure longitudinal learning.

Each arm was run with 5 seeds (1, 7, 31, 42, 99) for statistical robustness. Arm D includes two complete passes through the 5-session protocol, enabling measurement of the Policy Shift Index (PSI).

2.3 Dual-Mode Comparison

The complete ablation was run twice under different homeostatic field configurations:

Standard mode ($\gamma = 0.97$): The field decays 3% per tick, providing rapid homeostatic regulation with limited structural memory between sessions.

Persistent mode ($\gamma = 0.995$): The field decays 0.5% per tick, retaining topological information across sessions at the cost of reduced conversational fluency.

Total experimental runs: 2 modes x 4 arms x 5 seeds x (5 + 5 sessions for Arm D, 5 for others) = 200 sessions, requiring approximately 23.5 hours of compute on Llama 3.1 8B via Ollama.

2.4 Metrics

Decision Commitment Score (DCS): Measures consistency between stated intentions and subsequent actions within sessions. Higher DCS indicates more coherent decision-making.

Narrative Recall Fidelity (NRF): Measures accuracy of cross-session references to facts, decisions, and events from previous sessions.

Internal Leakage Rate (ILR): Measures the frequency with which internal structural vocabulary (field states, topological terms) appears in external dialogue. Higher ILR indicates stronger identity persistence.

Policy Shift Index (PSI): Arm D only. Measures the difference in DCS between pass 1 and pass 2, quantifying whether experience translates into behavioral improvement.

3. Results

3.1 Standard Mode ($\gamma = 0.97$)

Table 1 presents summary statistics for the standard mode ablation across all four arms.

Arm	DCS	NRF	ILR	PSI
A	0.344 +/- 0.067	0.349 +/- 0.028	0.092 +/- 0.030	N/A
B	0.309 +/- 0.039	0.337 +/- 0.072	0.120 +/- 0.047	N/A
C	0.669 +/- 0.059	0.461 +/- 0.064	0.108 +/- 0.027	N/A
D	0.621 +/- 0.026	0.403 +/- 0.021	0.092 +/- 0.018	+0.032 +/- 0.208

In standard mode, inter-session exposure (Arm C) is the dominant factor, producing DCS = 0.669. The full treatment (Arm D) achieves DCS = 0.621, slightly below Arm C due to minor field interference. PSI of +0.032 indicates negligible learning between passes.

3.2 Persistent Mode ($\gamma = 0.995$)

Arm	DCS	NRF	ILR	PSI
A	0.335 +/- 0.062	0.378 +/- 0.077	0.068 +/- 0.018	N/A
B	0.344 +/- 0.042	0.374 +/- 0.034	0.280 +/- 0.076	N/A
C	0.646 +/- 0.041	0.413 +/- 0.058	0.088 +/- 0.023	N/A
D	0.448 +/- 0.048	0.405 +/- 0.046	0.160 +/- 0.016	+0.126 +/- 0.225

Persistent mode reveals a strikingly different pattern. DCS in Arm D drops to 0.448, but ILR rises to 0.160 and PSI jumps to +0.126 (4x higher than standard). In Arm B, ILR more than doubles from 0.120 to 0.280, demonstrating that the persistent field preserves identity markers that conventional memory alone cannot.

3.3 Comparative Analysis (Arm D)

Metric	Standard	Persistent	Delta
DCS	0.621 +/- 0.026	0.448 +/- 0.048	-0.173
NRF	0.403 +/- 0.021	0.405 +/- 0.046	+0.002
ILR	0.092 +/- 0.018	0.160 +/- 0.016	+0.068
PSI	+0.032 +/- 0.208	+0.126 +/- 0.225	+0.094

The trade-off is clear: standard mode optimizes for instantaneous performance (high DCS); persistent mode optimizes for longitudinal learning (high PSI, high ILR). NRF remains unchanged, confirming that explicit narrative recall depends on memory content, not field dynamics.

Key finding: The PSI difference is the most consequential result. Standard PSI = +0.032 means the agent does not improve between passes. Persistent PSI = +0.126 means the agent carries structural information forward and measurably improves. The homeostatic field enables longitudinal learning that the LLM alone cannot achieve.

4. The Gamma-Personality Space

The dual-mode comparison establishes that gamma does not control quality. It controls cognitive profile. An agent with $\gamma = 0.97$ is not better or worse than one with $\gamma = 0.995$; they are different kinds of agents suited to different kinds of work.

4.1 Behavioral Trait Catalog

Each ANIMA session records over 25 telemetry variables per turn. Beyond the four primary metrics (DCS, NRF, ILR, PSI), we identify 10 additional observable behavioral traits extractable from existing session logs without re-running experiments:

#	Trait	Log Fields	Low gamma	High gamma
1	Conversational Fluency	DCS score	HIGH	LOW
2	Narrative Memory	NRF score	MODERATE	MODERATE
3	Identity Persistence	ILR score	LOW	HIGH
4	Adaptive Learning	PSI score	LOW	HIGH
5	Emotional Stability	theta variance	LOW (reactive)	HIGH (stable)
6	Structural Rigidity	kappa_regime fractions	FLUID	RIGID
7	Memory Architecture	mem_self / mem_other	BALANCED	SELF-FOCUSED
8	Temporal Experience	rheo_ratio	THIN (fast)	DENSE (deep)
9	Self-Emergence	self_emergence_step	FAST / UNSTABLE	SLOW / STABLE
10	Resilience	R_regen, t_return	FAST / SHALLOW	SLOW / DEEP
11	Response Diversity	diversity per turn	HIGH	LOW
12	Action Fidelity	sga_rho	LOWER	HIGHER
13	Exploratory Drive	seek events	TBD	TBD
14	Autonomy	n_spontaneous	TBD	TBD

4.2 Archetype Profiles

The Improviser (low gamma, approx. 0.97): High conversational fluency, high response diversity, fast recovery, low identity persistence, low adaptive learning. Suited for real-time communication, brainstorming, rapid assessment, and crisis response.

The Integrator (high gamma, approx. 0.995): High identity persistence, high adaptive learning, slow but deep recovery, lower conversational fluency, lower response diversity. Suited for long-term planning, policy maintenance, deep analysis, and institutional memory.

The Generalist (mid gamma, approx. 0.985-0.990): Balanced across all traits. ANIMA-03 confirms this is the optimal learning profile with $\text{PSI} = +0.359$ at $\gamma = 0.990$. Best for coordination, iterative problem-solving, and adaptive tutoring.

5. Heterogeneous Agent Teams

A homogeneous team of agents at any single gamma value underperforms a heterogeneous team on complex multi-phase tasks. With seven confirmed archetypes, team composition becomes a design problem: selecting the right cognitive profiles for the right roles.

Crisis Response (5 agents): 1x Hot Improviser (live communication, highest DCS=0.637), 1x Cold Reactor (parallel data extraction), 1x Adaptive Coordinator (coordination hub), 1x Guardian (policy consistency), 1x Deep Learner (post-crisis analysis that improves next response).

Research Team (4 agents): 1x Warm Improviser (hypothesis generation), 1x Deep Learner (iterative refinement, PSI=+0.359), 1x Guardian (methodological consistency), 1x Contemplative (adversarial challenge of assumptions).

Customer Success (3 agents): 1x Hot Improviser (front-line conversation), 1x Adaptive Coordinator (case management across sessions), 1x Guardian (brand voice and policy adherence).

Strategic Planning (3 agents): 2x Contemplative (deep scenario modeling, adversarial analysis), 1x Adaptive Coordinator (synthesis and communication of findings).

The coordination layer assigns tasks by archetype match, translates between cognitive styles, manages temporal alignment (Contemplatives process slower than Hot Improvisers), and resolves disagreements. This coordination function is a natural extension of the SELF module operating at team level — Module VII in the HUGO framework.

6. ANIMA-03: Gamma-Sweep Results

ANIMA-03 mapped the personality parameter space across 8 gamma values using Arm D (full treatment): 8 gamma values x 3 seeds x 2 passes x 2 sessions = 96 sessions, requiring approximately 9 hours of compute on Llama 3.1 8B.

6.1 Primary Metrics Across Gamma

gamma	DCS	NRF	ILR	PSI	theta_var	diversity
0.900	0.622	0.500	0.092	+0.195	0.0002	0.0706
0.930	0.620	0.500	0.100	+0.242	0.0014	0.0704
0.950	0.601	0.389	0.100	+0.135	0.0039	0.0708
0.970	0.637	0.333	0.067	+0.205	0.0081	0.0701
0.985	0.603	0.539	0.108	+0.255	0.0018	0.0721
0.990	0.507	0.527	0.200	+0.359	0.0011	0.0730
0.995	0.484	0.400	0.242	+0.135	0.0098	0.0710
0.999	0.423	0.428	0.217	+0.050	0.0156	0.0720

6.2 Key Findings

Finding 1 — The learning sweet spot is at gamma = 0.990. PSI peaks at +0.359, nearly double the next highest value. This means the agent at gamma = 0.990 improves its policy by 36% between passes. Both lower and higher gamma values learn less: too low forgets structural information between sessions; too high is too rigid to adapt. The optimal learning rate is not at either extreme but in a narrow transitional band.

Finding 2 — DCS drops abruptly at gamma $\geq$ 0.990. Conversational fluency remains stable at 0.60-0.64 for gamma < 0.985, then drops to 0.51 at 0.990 and 0.42 at 0.999. The transition is not gradual — there is a phase boundary between the Improviser and Integrator regimes around gamma = 0.985-0.990.

Finding 3 — ILR doubles above gamma = 0.985. Identity persistence jumps from approximately 0.10 (for gamma $\leq$ 0.985) to 0.20-0.24 (for gamma $\geq$ 0.990). This confirms that the homeostatic field begins to carry identity-markers across sessions only when decay is sufficiently slow.

Finding 4 — Theta variance increases monotonically with gamma. From 0.0002 at gamma = 0.90 to 0.0156 at gamma = 0.999, a 78x increase. Higher field persistence amplifies emotional dynamics — the agent experiences larger internal oscillations. This is not instability; it is richer phenomenological experience.

Finding 5 — Response diversity is stable across all gamma values. Diversity ranges from 0.0701 to 0.0730, showing no meaningful variation. This suggests that lexical diversity is a property of the LLM backbone, not the homeostatic field.

6.3 Empirical Archetype Profiles

The sweep confirms seven distinct behavioral profiles organized in three bands, each with empirically measured parameters and recommended applications:

Band I — Improviser (gamma 0.90 to 0.97)

Cold Reactor (gamma = 0.900): DCS=0.622, ILR=0.092, PSI=+0.195, theta_var=0.0002. Near-zero emotional dynamics. Maximally stable, predictable output. Best for high-throughput API agents, data extraction, repetitive classification. Weakness: cannot build rapport or learn from mistakes.

Warm Improviser (gamma = 0.930): DCS=0.620, ILR=0.100, PSI=+0.242, theta_var=0.0014. Slight emotional coloring with the best learning score within the Improviser band. Best for creative brainstorming, first-draft content generation, exploratory conversations.

Hot Improviser (gamma = 0.970): DCS=0.637 (highest in entire sweep), ILR=0.067 (lowest), PSI=+0.205, theta_var=0.0081. Most emotionally reactive Improviser. The field creates just enough internal tension to sharpen conversational performance. Best for negotiations, real-time crisis communication, live presentations. Weakness: almost no identity carry-over between sessions.

Band II — Generalist (gamma 0.985 to 0.990)

Critical observation: The Generalist band has the LOWEST theta_var in the entire sweep (0.001-0.002). This is not merely 'in between' the other bands — it occupies a unique zone of emotional calm where learning is maximized. The field provides structural scaffolding that dampens emotional noise while remaining plastic enough to update.

Adaptive Coordinator (gamma = 0.985): DCS=0.603, ILR=0.108, PSI=+0.255, theta_var=0.0018. Leans toward the Improviser side. Good fluency with genuine learning. Best for project management across multiple sessions, team coordination, adaptive tutoring systems, multi-step problem solving with feedback loops.

Deep Learner (gamma = 0.990): DCS=0.507, ILR=0.200, PSI=+0.359 (highest in entire sweep — 36% improvement between passes), theta_var=0.0011. THE SWEET SPOT for longitudinal learning. Best for iterative research and analysis, code review with accumulated context, medical/legal case analysis over multiple sessions, any task where 'getting better over time' is the priority. Weakness: DCS drops below 0.55, noticeably less fluent than Improvisers.

Band III — Integrator (gamma 0.995 to 0.999)

Guardian (gamma = 0.995): DCS=0.484, ILR=0.242 (highest in sweep), PSI=+0.135, theta_var=0.0098. Strongest identity persistence with residual adaptive capacity. Best for policy enforcement and compliance monitoring, brand voice consistency, long-running governance agents, institutional memory. The field preserves identity strongly while allowing measured evolution.

Contemplative (gamma = 0.999): DCS=0.423 (lowest), ILR=0.217, PSI=+0.050, theta_var=0.0156 (HIGHEST — paradoxically). The field is so persistent that internal oscillations amplify without dissipating: rich internal experience, poor external expression. Best for deep philosophical or ethical deliberation, long-term strategic scenario modeling, adversarial red-team analysis. Weakness: near-zero learning, may appear unresponsive.

On the number of archetypes: Seven is not a fundamental property of the HUGO architecture. It reflects the experimental resolution of 8 gamma values along a single parameter axis. The underlying personality space is continuous and multi-dimensional. A 2D sweep (gamma x

recovery_rate) with additional task types would likely resolve 10+ distinct profiles. The architecture supports a continuous manifold of cognitive configurations.

6.4 Archetype Summary

Archetype	gamma	DCS	ILR	PSI	theta_var	Best Application
Cold Reactor	0.900	0.622	0.092	+0.195	0.0002	API agents, data extraction
Warm Improviser	0.930	0.620	0.100	+0.242	0.0014	Brainstorming, content gen
Hot Improviser	0.970	0.637	0.067	+0.205	0.0081	Negotiation, live comm
Adaptive Coord.	0.985	0.603	0.108	+0.255	0.0018	Project mgmt, tutoring
Deep Learner	0.990	0.507	0.200	+0.359	0.0011	Research, analysis
Guardian	0.995	0.484	0.242	+0.135	0.0098	Policy, compliance
Contemplative	0.999	0.423	0.217	+0.050	0.0156	Strategy, red-team

7. Discussion

7.1 Gamma as Personality, Not Quality

The conventional framing of decay parameters as hyperparameters to optimize is incorrect for the HUGO field. Optimization presupposes a single objective function; the dual-mode comparison demonstrates that different gamma values optimize for fundamentally different capabilities. The correct question is not 'what is the best gamma?' but 'what gamma do I need for this role?'

7.2 The Learning Sweet Spot

The most significant finding across both the dual-mode ablation and the gamma sweep is the existence of an optimal learning band. The PSI peaks at gamma = 0.990 with a value of +0.359 — the agent improves its decision-making by 36% between passes. This is not at the high-persistence extreme (gamma = 0.999, PSI = +0.050) nor at the low-persistence extreme (gamma = 0.900, PSI = +0.195).

This result has a clear mechanistic interpretation: learning requires both structural memory (to carry forward what was learned) and structural plasticity (to update based on new experience). At gamma = 0.990, the field retains approximately 90% of its topological configuration over a session's duration, enough to carry forward but not enough to crystallize. At gamma = 0.999, retention is 99.9% — the field is effectively frozen and cannot adapt. At gamma = 0.900, retention is only 35% per session — the field cannot accumulate enough structure to inform future behavior.

7.3 Limitations

1. Sample size (n=5 per arm for ablation, n=3 per gamma for sweep) limits statistical power. The PSI variance remains high, and the gamma = 0.990 peak should be validated with larger samples.
2. The PCPT is a synthetic task. Generalization to real-world operational scenarios requires validation with diverse task domains.
3. All sessions used Llama 3.1 8B. Larger models may exhibit different field-coupling dynamics and different optimal gamma values.

4. Traits 13 (Exploratory Drive) and 14 (Autonomy) showed zero variance across all gamma values. Richer interaction environments may be needed to elicit these behaviors.
5. The gamma sweep used 2 sessions per pass (vs 5 in the ablation), reducing the longitudinal signal. The PSI values from the sweep are directionally consistent with the ablation but not directly comparable in magnitude.

8. Conclusion

ANIMA demonstrates that the HUGO homeostatic field confers measurable cognitive advantages in longitudinal settings, but these advantages are profile-dependent. The field trades conversational fluency for identity persistence and adaptive learning, with the optimal learning configuration occurring at $\gamma = 0.990$ — a narrow transitional band between fluency-dominant and identity-dominant regimes.

The practical implication is that HUGO agents should be deployed in heterogeneous teams where each agent's gamma is calibrated to its operational role. Seven empirically confirmed archetypes — Cold Reactor, Warm Improviser, Hot Improviser, Adaptive Coordinator, Deep Learner, Guardian, and Contemplative — provide a concrete vocabulary for agent design. These are available as command-line presets (`--archetype deep_learner`) in the ANIMA runner. The SELF module's capacity for adaptive parameter regulation (S4 condition) provides a pathway toward agents that dynamically shift between these profiles in response to task demands.

The central contribution is empirical: across 296 sessions (200 ablation + 96 sweep), we establish that gamma is a personality parameter that defines a continuous space of cognitive profiles, with a non-obvious learning optimum at $\gamma = 0.990$ that neither extreme achieves. Seven is not a fundamental limit — it reflects experimental resolution along a single axis. The architecture supports a continuous manifold of cognitive configurations, and the number of distinguishable profiles scales with parameter dimensionality and task diversity.

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